FileVault Full-Disk Encryption & Key Escrow/Recovery for macOS (Intune)

Implementation Solution Guide

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Contents

Executive Summary — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune) 4

What you get . . . 4

What this delivers . . . 4

The standard deployment model . . . 4

Outcome . . . 4

1. Solution Overview — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune) 4

What FileVault is . . . 5

How it works on modern Macs (Apple Silicon and Intel T2) . . . 5

Personal vs institutional recovery keys . . . 5

The recovery model . . . 5

What you can require through Intune . . . 6

Benefits . . . 6

Limitations (be honest with buyers) . . . 6

The standard deployment model . . . 6

2. Requirements Guide — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune) 7

2.1 Intune licensing . . . 7

2.2 Apple MDM Push (APNs) certificate . . . 7

2.3 macOS enrollment configured (and enrolled Macs) . . . 7

2.4 Supported hardware — Apple Silicon or Intel T2 . . . 7

2.5 SecureToken / bootstrap token available . . . 8

2.6 Microsoft Entra ID groups (pilot and production) . . . 8

2.7 A pilot (change control) . . . 8

Checklist . . . 8

Global Requirements — Intune Solutions (category-level) 9

G1. Microsoft Entra ID tenant . . . 9

G2. Microsoft Intune subscription . . . 9

G3. Administrative roles . . . 9

G4. Device enrollment in Intune . . . 9

G5. Pilot security group . . . 9

G6. Supported Windows baseline . . . 9

G7. Network access to Microsoft cloud endpoints . . . 9

3. Implementation Guide — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune) 10

Step 1 — Create the device groups . . . 10
Step 2 — Confirm SecureToken / bootstrap token . . . 10
Step 3 — Create the FileVault disk-encryption policy . . . 10
Step 4 — Configure FileVault settings (escrow + hide + rotate) . . . 10
Step 5 — Scope tags (optional) and Assign to the pilot . . . 11
Step 6 — Apply on a pilot Mac . . . 11
Step 7 — Validate the pilot (escrow + recovery) . . . 11
Step 8 — Expand to production . . . 12
5. Testing Guide — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune) 12
Test 1 — Policy applied . . . 12
Test 2 — FileVault is on and the volume is encrypted . . . 12
Test 3 — SecureToken confirmed for the FileVault user . . . 12
Test 4 — Personal recovery key escrowed to Intune . . . 13
Test 5 — Recovery-key retrieval works (Graph) . . . 13
Test 6 — Recovery unlock + rotation . . . 13
Test 7 — Compliance reflects encryption . . . 13
Test 8 — Fleet report . . . 13
Pass/fail gate . . . 13
6. Troubleshooting Guide — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune) 13
A. FileVault won't enable . . . 14
B. Recovery key not escrowed (critical) . . . 14
C. SecureToken / bootstrap-token problems . . . 14
D. Recovery key won't unlock / rotation issues . . . 15
E. Diagnostics quick reference (run on the Mac) . . . 15
F. When to escalate . . . 15
7. Best Practices — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune) 15
8. Deliverables Summary — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune) 16
Documentation . . . 16
PowerShell scripts . . . 16
What the buyer receives . . . 17
Outcome . . . 17
Appendix A — PowerShell Scripts 18

Executive Summary — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune)

White-label · Intune Solutions

A MacBook left in a taxi is only a breach if its drive is readable. **FileVault** encrypts the whole disk so a lost or stolen Mac exposes nothing — but only if it is **enforced, escrowed, and recoverable**. Managed through Microsoft Intune, this solution turns FileVault on by policy, **escrows every personal recovery key to Intune** (and rotates it), and handles the **SecureToken / bootstrap-token** enablement that FileVault depends on — so data is protected and the help desk can always recover a locked-out Mac.

This is a complete, **deploy-it-today** package for implementing FileVault encryption and recovery-key escrow with Microsoft Intune, written for an administrator with no prior FileVault- or SecureToken-management experience.

What you get

- A plain-language **Solution Overview** of how FileVault + SecureToken work on Apple Silicon/T2, personal vs institutional recovery keys, Intune escrow, and the recovery model.
- A **Requirements Guide** with verify / implement-if-missing for every prerequisite (enrollment, disk-encryption policy, SecureToken/bootstrap token, Entra groups, a pilot).
- An **Implementation Guide** of precise, click-by-click Intune steps in the required order.
- Five **PowerShell scripts** for readiness discovery, per-device validation, fleet + recovery-key reporting, safe remediation, and pilot/target group building.
- **Testing** and **Troubleshooting** guides for confident rollout, plus **Best Practices** for key rotation and recovery.

What this delivers

An Intune **FileVault disk-encryption policy** that enables FileVault on the target Macs, **escrows the personal recovery key to Intune**, **hides the key from the user**, and **rotates** it — with full encryption reporting and a proven recovery workflow.

The standard deployment model

Assess → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

Fleet-wide FileVault encryption with escrowed, rotatable recovery keys — device-loss data protection and a compliance cornerstone — deployable by a single administrator and validated before rollout.

1. Solution Overview — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune)

What FileVault is

FileVault is Apple's built-in full-disk (whole-volume) encryption for macOS. When enabled, the data on the Mac's internal drive is encrypted with **XTS-AES 128** and is unreadable without a credential that can unlock it. Managed through Microsoft Intune, FileVault is **enforced by policy**, and every **personal recovery key is escrowed to Intune** so an authorized admin can recover a locked-out Mac.

How it works on modern Macs (Apple Silicon and Intel T2)

On **Apple Silicon** Macs and **Intel Macs with the Apple T2 Security Chip**, the internal storage is *always* hardware-encrypted at the controller level. FileVault adds the crucial second layer: it ties decryption of the volume to a **user credential**, so the data is not readable simply by having the hardware. Turning FileVault on is therefore what makes the built-in encryption meaningful for device-loss protection — without it, the drive can be unlocked without a password.

FileVault depends on a macOS feature called **SecureToken**:

1. A **SecureToken** is a wrapped cryptographic key granted to a user account. Only a user who

holds a SecureToken can be enabled for FileVault (added to the list of users allowed to unlock the disk at boot).

1. On **Automated Device Enrollment (ADE/DEP)** Macs, Intune deploys a **bootstrap token** — a

token escrowed to MDM that lets the management channel grant SecureToken to accounts (including mobile/Entra accounts) without an existing SecureToken admin present. **This is why the bootstrap token matters: without it, a freshly provisioned ADE Mac may have no SecureToken user, and FileVault cannot be enabled.**

1. With a SecureToken-enabled user present, the Intune **FileVault disk-encryption policy** turns

FileVault on. On next login the user is prompted (or, where supported, enablement is deferred to the next sign-in) and encryption begins.

Personal vs institutional recovery keys

FileVault supports two recovery-key models. Intune uses the **personal recovery key** model:

• **Personal recovery key (what Intune manages):** a unique 24-character key is generated **per

device **during enablement and** escrowed to Intune**. It is visible to admins in the Intune admin center under the device's** Recovery keys**, and retrievable through Microsoft Graph. Intune can also** rotate** the personal recovery key after it has been used or on a schedule.

• **Institutional recovery key:** a single organization-wide key backed by a certificate/keychain

that unlocks any Mac. Intune's FileVault profile is built around the **personal** key model; institutional keys are a legacy, higher-risk pattern (one key unlocks the whole fleet) and are not the recommended Intune approach. This solution uses **personal keys escrowed and rotated by Intune**.

The recovery model

When a user forgets their password or a Mac is locked out:

1. The admin looks up the device in Intune → device → **Recovery keys** (or retrieves it through Microsoft Graph — see `MACFV-Report.ps1`).

1. At the FileVault login/recovery screen, the personal recovery key unlocks the volume.

2. The user resets their password; Intune **rotates** the personal recovery key so the used key is retired and a fresh one is escrowed.

What you can require through Intune

• **Enable FileVault** on the target Macs (disk-encryption policy, platform macOS, profile **FileVault**).

• **Escrow the personal recovery key to Intune** and **hide the recovery key from the user**.

• **Rotate** the personal recovery key (after use / on a schedule).

• **Defer enablement** until the user next logs out/in, with an optional prompt limit.

• Deploy the **bootstrap token** (via ADE enrollment) so SecureToken can be granted.

Benefits

• **Device-loss protection** — a lost or stolen Mac exposes no readable data.

• **Always recoverable** — personal keys escrowed to Intune; the help desk can always unlock a Mac.

• **Rotatable keys** — a used recovery key can be retired automatically.

• **Compliance cornerstone** — satisfies "require encryption" in device compliance / Conditional Access.

Limitations (be honest with buyers)

• **FileVault needs a SecureToken-enabled user.** On non-ADE (manually enrolled) Macs there may be no bootstrap token, so SecureToken must be granted by an existing SecureToken user — plan for this.

• **Enablement often requires the user to sign in** (FileVault is enabled at login on the personal key model); it is not always fully silent.

• The **first full encryption takes time**; plan pilot timing.

• FileVault's encryption method is **XTS-AES 128** (Apple's implementation) — this is expected and is not the same knob as Windows BitLocker's selectable method.

The standard deployment model

> ***Confirm SecureToken/bootstrap-token → create the FileVault disk-encryption policy (escrow + rotate) → assign to a pilot → validate encryption + key escrow + recovery → expand.*** *The repeatable pattern is:* ***Assess → Configure → Pilot → Validate → Expand.***

2. Requirements Guide — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune)

Verify **every** requirement before configuring. Each entry: **Description · Why required · How to verify · How to implement if missing.** Category-wide prerequisites are in `global-requirements-intune-solutions.md`.

Object names used throughout this solution (reuse them verbatim in Intune):

- Disk-encryption policy: `FileVault – Pilot` (and later `FileVault – Production`)
- Pilot device group: `FileVault – Pilot Devices`
- Production device group: `FileVault – Production Devices`
- (If used) macOS enrollment profile / bootstrap-token source: your Automated Device Enrollment profile

2.1 Intune licensing

Description: An Intune-bearing license (Plan 1 is sufficient for the macOS disk-encryption policy). **Why required:** Intune delivers the FileVault disk-encryption policy, escrows the recovery key, and reports encryption status. **How to verify:** Microsoft 365 admin center → **Billing** → **Licenses**. **How to implement if missing:** Assign M365 E3/E5 or Business Premium (Intune included), or standalone Intune Plan 1.

2.2 Apple MDM Push (APNs) certificate

Description: A valid Apple Push Notification service (APNs) certificate configured in Intune. **Why required:** No Apple device — including a Mac — can be managed by Intune without an APNs certificate; no policy reaches the Mac until it exists. **How to verify:** Intune → **Devices** → **Enrollment** → **Apple** → **Apple MDM Push certificate**; confirm it is present and **not expired**. Run `MACFV-Discovery.ps1`. **How to implement if missing:** Intune → **Devices** → **Enrollment** → **Apple** → **Apple MDM Push certificate** → **download the CSR**, create the certificate at `https://identity.apple.com` with a long-lived Apple ID, upload it back, and record the renewal date.

2.3 macOS enrollment configured (and enrolled Macs)

Description: A working macOS enrollment method so target Macs are enrolled and manageable — Automated Device Enrollment (ADE/DEP), Account-driven enrollment, or Company Portal. **Why required:** The FileVault policy and its escrow/reporting apply only to enrolled Macs. **How to verify:** Intune → **Devices** → **macOS** → **macOS devices**; the Mac appears as managed. Run `MACFV-Discovery.ps1`. **How to implement if missing:** Configure **Automated Device Enrollment** (Apple Business/School Manager token + an enrollment profile) for corporate Macs, or **Account-driven**/Company Portal enrollment for user-owned Macs; then enrol the devices.

2.4 Supported hardware — Apple Silicon or Intel T2

Description: Target Macs are **Apple Silicon** or **Intel Macs with the Apple T2 Security Chip** (older Intel Macs still support FileVault but lack the hardware layer and bootstrap-token behaviour). **Why required:**

SecureToken/bootstrap-token behaviour and the always-on hardware encryption described in the Overview depend on Apple Silicon / T2. **How to verify:** On a Mac, **Apple menu** → **About This Mac** (Apple Silicon shows a "Chip"; Intel T2 Macs list the T2 under **System Report** → **Controller**). Or Intune → device → **Hardware**. **How to implement if missing:** For unsupported older hardware, FileVault still encrypts but plan SecureToken enablement manually and expect no bootstrap-token flow.

2.5 SecureToken / bootstrap token available

Description: At least one account on each Mac can be granted **SecureToken**, and (on ADE Macs) a **bootstrap token** is escrowed to Intune. **Why required:** FileVault **cannot be enabled** for a user without a SecureToken. On ADE Macs the bootstrap token lets Intune grant SecureToken to accounts; without it a fresh ADE Mac may have no SecureToken user. **How to verify:** On a Mac, `sysadminctl -secureTokenStatus <username>` shows **ENABLED** for the target user; `profiles status -type bootstraptoken` shows the bootstrap token is escrowed to the server. Run `MACFV-Discovery.ps1` for the tenant/enrollment signals. **How to implement if missing:** Enroll corporate Macs via **Automated Device Enrollment** so the bootstrap token is generated and escrowed at setup. For already-enrolled Macs, run `sudo profiles install -type bootstraptoken` from a SecureToken-enabled admin, or grant SecureToken with `sysadminctl -secureTokenOn` from an existing SecureToken admin.

2.6 Microsoft Entra ID groups (pilot and production)

Description: Two Entra ID security groups: `FileVault – Pilot Devices` (a small representative sample of Macs) and `FileVault – Production Devices` (the rest of the Mac fleet). **Why required:** FileVault is piloted before fleet rollout because enablement affects user login and the first encryption takes time. **How to verify:** Entra → **Groups**; both groups exist. Use `MACFV-Set-ScopeTargeting.ps1` to stage membership. **How to implement if missing:** Entra → **Groups** → **New group** → **Security**; add representative pilot Macs to one and the remaining Macs to the other.

2.7 A pilot (change control)

Description: A documented approval step and a small pilot of representative Macs before fleet rollout, with a rollback plan. **Why required:** Enabling FileVault is a production change that touches user login; validate escrow and recovery on a pilot before scaling. **How to verify:** You have a change record (who approved, what changed, rollback step) and a populated `FileVault – Pilot Devices` group. **How to implement if missing:** Adopt a simple change record and populate the pilot group with a representative sample.

Checklist

- Intune licensing assigned
- Apple MDM Push (APNs) certificate present and not expired — `MACFV-Discovery.ps1`
- macOS enrollment configured (ADE / Account-driven / Company Portal); Macs enrolled
- Target Macs are Apple Silicon or Intel T2
- SecureToken available / bootstrap token escrowed (ADE) — `MACFV-Discovery.ps1`
- `FileVault – Pilot Devices` and `FileVault – Production Devices` groups created

- Pilot + change-control/approval process agreed

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation Guide — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune)

Follow these steps **in order**, in the **Microsoft Intune admin center** (`https://intune.microsoft.com`).

> ***Model:*** *A macOS **FileVault** disk-encryption policy that enables FileVault, **escrows the personal recovery key to Intune**, **hides the key from the user**, and **rotates** it — assigned to a pilot group first, then expanded. FileVault depends on **SecureToken**, so confirm the bootstrap token first.*

Step 1 — Create the device groups

1. In the Intune admin center go to **Groups** → **New group** (this opens Microsoft Entra).
2. **Group type:** Security. **Group name:** `FileVault – Pilot Devices`. Add a small, representative set of pilot Macs. **Create**.
1. Repeat to create `FileVault – Production Devices` for the rest of the Mac fleet (populate it later, at Step 8).

Step 2 — Confirm SecureToken / bootstrap token

1. For **ADE (Automated Device Enrollment)** Macs, the bootstrap token is generated and escrowed at setup. Confirm on a pilot Mac: `profiles status -type bootstraptoken` (should report the token escrowed to the server), and `sysadminctl -secureTokenStatus <username>` shows **ENABLED** for the account that will use FileVault.
1. Run `MACFV-Discovery.ps1` to confirm APNs, enrollment, and readiness at the tenant level.
2. **Gate:** Do not proceed until at least one account on each pilot Mac is SecureToken-enabled (see Requirements 2.5). Without SecureToken, FileVault will not enable.

Step 3 — Create the FileVault disk-encryption policy

1. Intune → **Endpoint security** → **Disk encryption** → **Create policy**.
2. **Platform: macOS**. **Profile: FileVault**. **Create**.
3. **Name:** `FileVault – Pilot`. (Optional) Description: note the pilot scope. **Next**.

Step 4 — Configure FileVault settings (escrow + hide + rotate)

On **Configuration settings**, set FileVault to enable, escrow the personal key, hide it from the user, and rotate it:

1. **Enable FileVault** = **Yes**.
2. **Escrow location description of personal recovery key** = a short message the user sees, e.g.

```
Your recovery key is stored securely by IT. Contact the Help Desk if you are locked out.
```

1. **Hide recovery key from user** = **Yes** (the personal recovery key is escrowed to Intune, not shown on screen).
1. **Personal recovery key rotation** = **Enabled** — set a rotation period (for example every **6 months**) so keys refresh over time.
1. **Number of times user can bypass** (deferral) = choose a small number (for example **3**) so enablement is enforced after a few logins rather than deferred indefinitely.
1. **Disable prompt at sign out** / defer-until-logout options = leave at the defaults unless you have a reason to change them. Click **Next**.

> ***Note:*** *Intune's FileVault profile uses the* ***personal recovery key*** *model (per-device key escrowed to Intune). It does not configure an institutional key — that legacy model is out of scope by design (see Overview).*

Step 5 — Scope tags (optional) and Assign to the pilot

1. **Scope tags** — add if you use RBAC scoping. **Next**.
2. **Assignments** → **Add groups** → `FileVault - Pilot Devices`. **Next** → **Review + create** → **Create**.

Step 6 — Apply on a pilot Mac

1. On a pilot Mac, sync policy (**System Settings** → **Privacy & Security** → **Profiles**, or force a check-in from Company Portal).
1. At the next **log out / log in**, the SecureToken-enabled user is prompted to enable FileVault (up to the bypass count you set). After the user signs in, **encryption begins**.

Step 7 — Validate the pilot (escrow + recovery)

1. Run `MACFV-Validation.ps1 -DeviceName "MAC-001"` → confirms Intune sees the device as **encrypted** and that a **personal recovery key is escrowed**.
1. In Intune → **Devices** → **macOS** → **<device>** → **Recovery keys**, confirm the **personal recovery key is present**.
1. **Recovery workflow test:** on the pilot Mac, at the login screen choose the "reset password /

recovery key" path, enter the escrowed personal recovery key retrieved from Intune (or via `MACFV-Report.ps1`), and confirm the volume unlocks. Then confirm Intune **rotates** the key (a new key appears; the used one is retired).

1. Run `MACFV-Report.ps1` → confirms encryption + escrow coverage across the pilot group.

Complete the **Testing Guide** (document 5).

> ***Pilot pass gate:*** *every pilot Mac is* ***encrypted****, has an* ***escrowed personal recovery key****, and the* ***recovery-key unlock + rotation*** *worked. Do not expand until this passes.*

Step 8 — Expand to production

1. Populate `FileVault – Production Devices` and add it to the `FileVault – Pilot` policy's **Assignments** (or clone the policy as `FileVault – Production`); roll out in waves.
1. For each wave, confirm **escrow for every device** — a Mac that encrypts without an escrowed key is a recovery risk. Use `MACFV-Report.ps1`.
1. If any Mac is encrypted but **not escrowed** or FileVault did not enable, run `MACFV-Remediation.ps1` (preview first; it defaults to `-WhatIf`).

Order of operations

`Device groups → Confirm SecureToken/bootstrap token → FileVault policy (enable + escrow + hide + rotate) → Assign to pilot → Apply at login → Validate encryption + escrow + recovery → Expand`

5. Testing Guide — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune)

Run after the pilot policy applies (Implementation Step 7) and after each production wave. Use `MACFV-Validation.ps1` for a single Mac and `MACFV-Report.ps1` for the fleet.

Test 1 — Policy applied

Do: Intune → **Endpoint security** → **Disk encryption** → `FileVault – Pilot` → **Device status**. **Expected:** Pilot devices show **Succeeded** (no Error/Conflict).

Test 2 — FileVault is on and the volume is encrypted

Do: On a pilot Mac run `fdesetup status` (expect `FileVault is On`) and `diskutil apfs list` (the data volume shows FileVault/encryption). Or run `MACFV-Validation.ps1 -DeviceName "MAC-001"`. **Expected:** FileVault **On**; the internal volume is encrypted.

Test 3 — SecureToken confirmed for the FileVault user

Do: On the pilot Mac run `sysadminctl -secureTokenStatus <username>`. **Expected:** The user is **ENABLED** for SecureToken (and appears in `fdesetup list`).

Test 4 — Personal recovery key escrowed to Intune

Do: Intune → **Devices** → **macOS** → **<device>** → **Recovery keys**; or run `MACFV-Validation.ps1`. **Expected:** A **personal recovery key** is present for the device in Intune.

Test 5 — Recovery-key retrieval works (Graph)

Do: Run `MACFV-Report.ps1 -DeviceName "MAC-001"` (retrieves the escrowed personal recovery key for that Mac through Microsoft Graph). **Expected:** The personal recovery key is returned for an authorized admin.

Test 6 — Recovery unlock + rotation

Do: At the pilot Mac's login/recovery screen, use the escrowed personal recovery key to unlock, reset the password, then re-check Intune → **Recovery keys**. **Expected:** The key unlocks the volume; Intune **rotates** the key (a new key is escrowed, the used one retired).

Test 7 — Compliance reflects encryption

Do: Intune → device → **Compliance** (with a "require encryption / FileVault" compliance rule for macOS). **Expected:** The encrypted Mac is **Compliant** for the encryption requirement.

Test 8 — Fleet report

Do: Run `MACFV-Report.ps1` (no `-DeviceName`). **Expected:** A breakdown of encrypted / not-encrypted Macs and a flag for any encrypted Mac **without an escrowed key**.

Pass/fail gate

Gate	Requirement
Pilot pass	Tests 1–6 pass for the pilot group (escrow confirmed and recovery/rotation verified for every device)
Wave pass	Tests 2, 4, 8 pass before the next wave (encrypted + escrowed, no unescrowed devices)

6. Troubleshooting Guide — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune)

Use `MACFV-Validation.ps1` / `MACFV-Discovery.ps1` for state and `MACFV-Remediation.ps1` for safe fixes (it defaults to `-WhatIf`).

A. FileVault won't enable

Symptom	Cause	Remediation
Policy "Succeeded" but FileVault stays Off	No SecureToken-enabled user on the Mac	Check `sysadminctl -secureTokenStatus <username>`; grant SecureToken from an existing SecureToken admin (`sysadminctl -secureTokenOn`), or use the bootstrap token on ADE Macs
ADE Mac has no SecureToken user	Bootstrap token not escrowed	Confirm `profiles status -type bootstraptoken`; install with `sudo profiles install -type bootstraptoken`; re-enroll via ADE if needed
User never prompted to enable	Deferral bypass count exhausted or not reached	Confirm the FileVault policy's bypass/deferral count; have the user log out/in

B. Recovery key not escrowed (critical)

Symptom	Cause	Remediation
FileVault On but no key in Intune → Recovery keys	Escrow/hide-key not configured, or the Mac encrypted before the policy	Confirm the policy sets **Hide recovery key from user = Yes**; run `MACFV-Remediation.ps1` to re-apply the policy (device sync) so Intune escrows a fresh key
Key missing for pre-existing FileVault Macs	Encrypted before Intune managed it	Re-apply the FileVault policy so a new personal recovery key is generated and escrowed (rotation); verify with `MACFV-Report.ps1`

C. SecureToken / bootstrap-token problems

- `sysadminctl -secureTokenStatus <user>` shows **DISABLED** → grant from a SecureToken admin

(`sysadminctl -secureTokenOn <user> -password - -adminUser <admin> -adminPassword -`).

- `profiles status -type bootstraptoken` shows not escrowed → run `sudo profiles install -type bootstraptoken`

from a SecureToken-enabled account (ADE Macs generate this at setup).

- Mobile/Entra account can't be FileVault-enabled → it lacks SecureToken; grant it via the bootstrap

token or from an existing SecureToken admin.

D. Recovery key won't unlock / rotation issues

• Escrowed key does not unlock at the login screen → confirm you retrieved the **current** key (rotation may have replaced it); re-check Intune → **Recovery keys** or `MACFV-Report.ps1`.

• Key not rotating after use → confirm **Personal recovery key rotation** is enabled in the policy; force a device sync so the policy re-evaluates.

E. Diagnostics quick reference (run on the Mac)

```
fdesetup status                              # FileVault On/Off, deferral state
fdesetup list                                # users enabled for FileVault
sysadminctl -secureTokenStatus <username>    # SecureToken ENABLED/DISABLED for a user
profiles status -type bootstraptoken         # bootstrap token escrowed to server?
diskutil apfs list                           # volume encryption state
```

From the management side, `MACFV-Report.ps1` retrieves the escrowed personal recovery key and reports fleet encryption + escrow coverage.

F. When to escalate

If a Mac is Apple Silicon/T2, is enrolled, has a SecureToken-enabled user and an escrowed bootstrap token, the FileVault policy shows **Succeeded**, and FileVault still won't enable or the key won't escrow after remediation, collect `fdesetup status`, `sysadminctl -secureTokenStatus`, and `profiles status -type bootstraptoken` output and open a Microsoft support case.

7. Best Practices — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune)

• **Confirm SecureToken before you enable FileVault.** FileVault cannot turn on for a user without a SecureToken. On ADE Macs, make sure the **bootstrap token** is escrowed so Intune can grant it.

• **Enroll corporate Macs via Automated Device Enrollment.** ADE generates and escrows the bootstrap token automatically, which is the cleanest path to SecureToken and FileVault.

• **Escrow the personal recovery key — always.** Set **Hide recovery key from user = Yes** so the key is escrowed to Intune, not left on screen. A Mac that encrypts without an escrowed key is a recovery disaster waiting to happen.

• **Use the personal recovery key model, not institutional.** A per-device key escrowed to Intune limits blast radius; an institutional key is one key that unlocks the entire fleet — avoid it.

• **Turn on key rotation.** Enable **personal recovery key rotation** so a key that has been used (e.g. during a recovery) is retired and a fresh one is escrowed. Rotate on a schedule too.

• **Pilot first.** Confirm enablement, escrow, and the recovery-key unlock on a small group before

fleet rollout — enablement touches user login and the first encryption takes time.

• **Plan enablement timing.** FileVault enables at login on the personal-key model; set a small deferral/bypass count so it is enforced after a few sign-ins rather than deferred forever.

• **Pair with compliance.** Use a macOS "require encryption / FileVault" compliance rule so unencrypted Macs are flagged (and blocked via Conditional Access).

• **Document the recovery procedure.** Make sure the help desk knows how to retrieve a personal recovery key (Intune device → **Recovery keys**, or `MACFV-Report.ps1`) and how the login-screen recovery flow works.

• **Report continuously.** Track encryption + escrow coverage with `MACFV-Report.ps1`; investigate any Mac that is encrypted but not escrowed.

• **Mind SecureToken on account changes.** Deleting or migrating the SecureToken-enabled account can strand FileVault — have a recovery key in hand before making account changes.

8. Deliverables Summary — FileVault Full-Disk Encryption & Key Escrow for macOS (Microsoft Intune)

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	FileVault + SecureToken on Apple Silicon/T2, personal vs institutional keys, Intune escrow, recovery model
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Intune Solutions)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered, click-by-click Intune steps
5. Testing Guide	Validation procedures with pass/fail gates
6. Troubleshooting Guide	Field issues and remediation (SecureToken / FileVault)
7. Best Practices	Field-proven recommendations (key rotation, recovery)

PowerShell scripts

Script	What it does
`MACFV-Discovery.ps1`	Tenant/device readiness: APNs certificate, enrolled Macs, macOS-version spread, existing FileVault/disk-encryption policies (Graph, read-only)

Script	What it does
`MACFV-Validation.ps1`	Per-device check: Intune sees the Mac as encrypted and a personal recovery key is escrowed (Graph, read-only)
`MACFV-Report.ps1`	Fleet encryption + escrow coverage, and retrieval of a device's escrowed personal recovery key (Graph, read-only)
`MACFV-Remediation.ps1`	Safe remediation of unencrypted/unescrowed Macs — re-applies the FileVault policy via device sync (defaults to -WhatIf)
`MACFV-Set-ScopeTargeting.ps1`	Builds/adjusts pilot or production group membership for staged rollout (defaults to -WhatIf)

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process following the model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

Every Mac encrypted with FileVault and every **personal recovery key escrowed and rotatable** in Intune — device-loss protection with no lock-out you can't recover, proven on a pilot before you scale.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

MACFV-Discovery.ps1

Purpose: Reports whether the tenant and its Macs are ready to receive a FileVault disk-encryption policy: that an Apple MDM Push (APNs) certificate exists and is not near expiry, how many managed Macs are enrolled, the macOS-version spread, how many Macs already report as encrypted, and whether any existing macOS disk-encryption/FileVault policies are present (potential conflicts). Read-only against Microsoft Graph (Intune). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\MACFV-Discovery.ps1 Requires the Microsoft Graph SDK and these read scopes (consented on first sign-in): DeviceManagementServiceConfig.Read.All, DeviceManagementConfiguration.Read.All, DeviceManagementManagedDevices.Read.All. EXPECTED RESULTS APNs certificate presence/expiry, count of enrolled Macs, a macOS-version breakdown, an encrypted-vs-not count, and a list of existing macOS disk-encryption policies to review before you build. PASS/FAIL readiness signals for the FileVault rollout. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param()

How to run: powershell -ExecutionPolicy Bypass -File .\MACFV-Discovery.ps1 Requires the Microsoft Graph SDK and these read scopes (consented on first sign-in): DeviceManagementServiceConfig.Read.All, DeviceManagementConfiguration.Read.All, DeviceManagementManagedDevices.Read.All. EXPECTED RESULTS APNs certificate presence/expiry, count of enrolled Macs, a macOS-version breakdown, an encrypted-vs-not count, and a list of existing macOS disk-encryption policies to review before you build. PASS/FAIL readiness signals for the FileVault rollout. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param()

MACFV-Remediation.ps1

Purpose: When MACFV-Validation / MACFV-Report shows a Mac that is not encrypted, or is encrypted but has NO escrowed personal recovery key, this issues an Intune device sync so the assigned FileVault disk-encryption policy re-evaluates and Intune escrows (or rotates to) a fresh personal recovery key. It does NOT change FileVault on the Mac directly and does NOT decrypt anything — Intune stays the single source of truth. On-device enablement still requires a SecureToken-enabled user (see Troubleshooting A/C). HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\MACFV-Remediation.ps1 Apply: powershell -ExecutionPolicy Bypass -File .\MACFV-Remediation.ps1 -WhatIf:$false Target one Mac: powershell -ExecutionPolicy Bypass -File .\MACFV-Remediation.ps1 -DeviceName "MAC-001" -WhatIf:$false Requires the Microsoft Graph SDK and DeviceManagementManagedDevices.Read.All + DeviceManagementManagedDevices.PrivilegedOperations.All (for the sync action + key read). EXPECTED RESULTS With -WhatIf (default): lists the at-risk Macs and the sync actions it WOULD issue. With -WhatIf:$false: issues a sync to each so the FileVault policy re-applies. Re-run MACFV-Validation.ps1 after a policy cycle. .NOTES White-label — the repeatable way. Defaults to -WhatIf (no change unless -WhatIf:$false). Read-only against Graph otherwise. #> [CmdletBinding(SupportsShouldProcess=$true, ConfirmImpact='High')] param([string]$DeviceName, [switch]$WhatIf = $true)

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\MACFV-Remediation.ps1 Apply: powershell -ExecutionPolicy Bypass -File .\MACFV-Remediation.ps1 -WhatIf:$false Target one Mac: powershell -ExecutionPolicy Bypass -File .\MACFV-Remediation.ps1 -DeviceName "MAC-001" -WhatIf:$false Requires the Microsoft Graph SDK and DeviceManagementManagedDevices.Read.All + DeviceManagementManagedDevices.PrivilegedOperations.All (for the sync action + key read). EXPECTED RESULTS With -WhatIf (default): lists the at-risk Macs and the sync actions it WOULD issue. With -WhatIf:$false: issues a sync to each so the FileVault policy re-applies. Re-run MACFV-Validation.ps1 after a policy cycle. .NOTES White-label — the repeatable way. Defaults to -WhatIf (no change unless -WhatIf:$false). Read-only against Graph otherwise. #> [CmdletBinding(SupportsShouldProcess=$true, ConfirmImpact='High')] param([string]$DeviceName, [switch]$WhatIf = $true)

MACFV-Report.ps1

Purpose: Two modes, both read-only against Microsoft Graph (Intune): 1. Fleet report (default): summarizes managed Macs by encryption state and flags any Mac reported as encrypted but with NO escrowed personal recovery key — the recovery risk. 2. Key retrieval: with -DeviceName, retrieves that Mac's escrowed personal FileVault recovery key for an authorized admin (the help-desk recovery workflow). HOW TO RUN Fleet: powershell -ExecutionPolicy Bypass -File .\MACFV-Report.ps1 -CsvPath C:\Temp\filevault.csv Key: powershell -ExecutionPolicy Bypass -File .\MACFV-Report.ps1 -DeviceName "MAC-001" Requires the Microsoft Graph SDK and DeviceManagementManagedDevices.Read.All + DeviceManagementManagedDevices.PrivilegedOperations.All (to read escrowed FileVault keys). EXPECTED RESULTS Fleet mode: an encrypted/not-encrypted breakdown, a count of encrypted-without-escrow Macs, and an optional per-device CSV. Key mode: the personal recovery key for that Mac. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. Treat retrieved keys as secrets. #> [CmdletBinding()] param([string]$DeviceName, [string]$CsvPath)

How to run: Fleet: powershell -ExecutionPolicy Bypass -File .\MACFV-Report.ps1 -CsvPath C:\Temp\filevault.csv Key: powershell -ExecutionPolicy Bypass -File .\MACFV-Report.ps1 -DeviceName "MAC-001" Requires the Microsoft Graph SDK and DeviceManagementManagedDevices.Read.All + DeviceManagementManagedDevices.PrivilegedOperations.All (to read escrowed FileVault keys). EXPECTED RESULTS Fleet mode: an encrypted/not-encrypted breakdown, a count of encrypted-without-escrow Macs, and an optional per-device CSV. Key mode: the personal recovery key for that Mac. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. Treat retrieved keys as secrets. #> [CmdletBinding()] param([string]$DeviceName, [string]$CsvPath)

MACFV-Set-ScopeTargeting.ps1

Purpose: Moves a Mac (device object) into the correct FileVault pilot or production Entra security group so the FileVault disk-encryption policy can be staged before the fleet. Defaults to -WhatIf so it only previews the membership change; re-run with -WhatIf:$false to commit. Never disables a policy or widens fleet targeting. Replace the sample values your-group-name / your-object-id. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\MACFV-Set-ScopeTargeting.ps1 -GroupName your-group-name -MemberObjectId your-object-id (preview only; add -WhatIf:$false to commit)

How to run: powershell -ExecutionPolicy Bypass -File .\MACFV-Set-ScopeTargeting.ps1 -GroupName your-group-name -MemberObjectId your-object-id (preview only; add -WhatIf:$false to commit)

MACFV-Validation.ps1

Purpose: Checks one Mac in Intune for two things: that it reports as encrypted, and that a personal FileVault recovery key is actually escrowed to Intune for it. Reads the managed-device record (isEncrypted) and attempts to read the escrowed FileVault key via Graph. Reports PASS/FAIL for encryption and for escrow. Read-only — it changes nothing on the Mac. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\MACFV-Validation.ps1 -DeviceName "MAC-001" Requires the Microsoft Graph SDK and DeviceManagementManagedDevices.Read.All + DeviceManagementManagedDevices.PrivilegedOperations.All (to read the escrowed FileVault key). EXPECTED RESULTS A PASS/FAIL line for encryption and for recovery-key escrow, and "DEVICE ENCRYPTED + ESCROWED" when both pass. Exit 0 = pass, 1 = encrypted but not escrowed, 2 = not found / not encrypted. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([Parameter(Mandatory=$true)][string]$DeviceName)

How to run: powershell -ExecutionPolicy Bypass -File .\MACFV-Validation.ps1 -DeviceName "MAC-001" Requires the Microsoft Graph SDK and DeviceManagementManagedDevices.Read.All + DeviceManagementManagedDevices.PrivilegedOperations.All (to read the escrowed FileVault key). EXPECTED RESULTS A PASS/FAIL line for encryption and for recovery-key escrow, and "DEVICE ENCRYPTED + ESCROWED" when both pass. Exit 0 = pass, 1 = encrypted but not escrowed, 2 = not found / not encrypted. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([Parameter(Mandatory=$true)][string]$DeviceName)